How-To Library

Informatica

Performance tuning guidelines for pushdown optimization using Google BigQuery V2 Connector

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Abstract

You can configure pushdown optimization in a mapping that reads from or writes to Google BigQuery. This article describes the various pushdown scenarios that you can configure to optimize the performance of Google BigQuery V2 Connector.

Supported Versions

- Informatica® Cloud Data Integration

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Overview

When you apply pushdown optimization, the task pushes transformation logic to the source or target database based on the optimization type you specify in the task properties. Data Integration translates the transformation logic into Google BigQuery commands to the Google BigQuery database. The database runs the Google BigQuery commands to process the transformations.

This document describes the various pushdown scenarios that you can configure and how you cen optimize the performance of a Google BigQuery V2 mapping. The case studies also describe the mapping scenarios in which you can gain maximum performance using source pushdown optimization.

The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of the pushdown using Google BigQuery V2 Connector might vary based on individual environments and other parameters even when you use the same data.

Pushdown optimization types

You can configure the following types of pushdown optimization to optimize your Google BigQuery V2 mapping:

Source-side pushdown optimization

When you run a task enabled with source pushdown optimization, the task pushes down as much as the transformation logic as possible to process in the source database. Source pushdown optimization does not include the Target transformations.

Full pushdown optimization

When you run a task enabled with full pushdown optimization, the task pushes as much of the transformation logic as possible to process in the target database.

Partial pushdown optimization

Partial pushdown uses a combination of source and full pushdown. When you run a mapping with full pushdown optimization, the mapping attempts to perform full pushdown in the beginning and pushes the source, target, and other transformations configured in the mapping to the target.

If all the transformations in the mapping are supported for pushdown optimization, the task pushes down the entire mapping logic to the target database. If there is an unsupported transformation or unsupported function in the mapping while attempting the pushdown, full pushdown is converted into source pushdown. In this case, the task employs partial pushdown optimization and pushes down the mapping logic to the source database until the point where the transformation is supported for pushdown optimization in the mapping.

Case studies

To obtain optimal performance, the mapping design plays a key role in the pushdown performance.

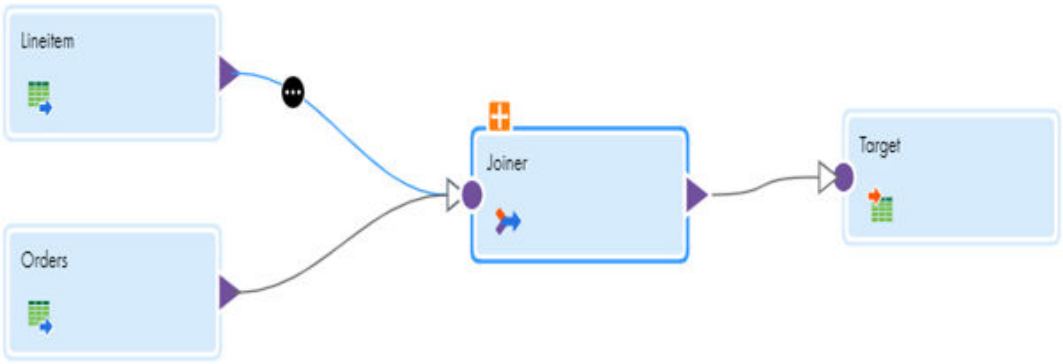
For example, to achieve optimal performance when you enable source pushdown optimization for a mapping that involves multiple transformations, you must design the mapping in such a way that heavy transformations, such as the Joiner transformation, must be placed further downstream closer to the Target transformation.

The case studies in this article show the results of mappings configured with full pushdown, source pushdown, and partial pushdown optimization.

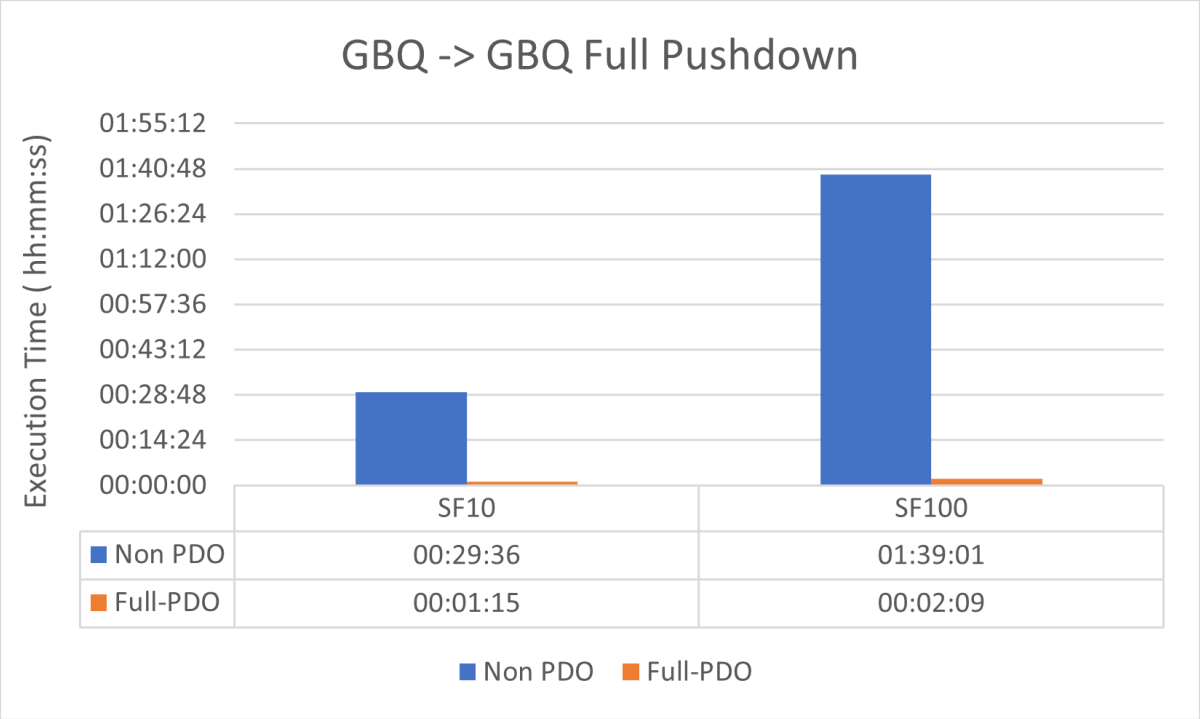
Case 1. Mapping with multiple sources and a single target

In this scenario, the mapping consists of two Google BigQuery tables as the source, a Joiner transformation, and a single Google BigQuery target. The mapping runs with full pushdown optimization.

The following figure shows the configured mapping enabled for full pushdown optimization:



The following figure shows the results when you run the mapping enabled with and without pushdown optimization:

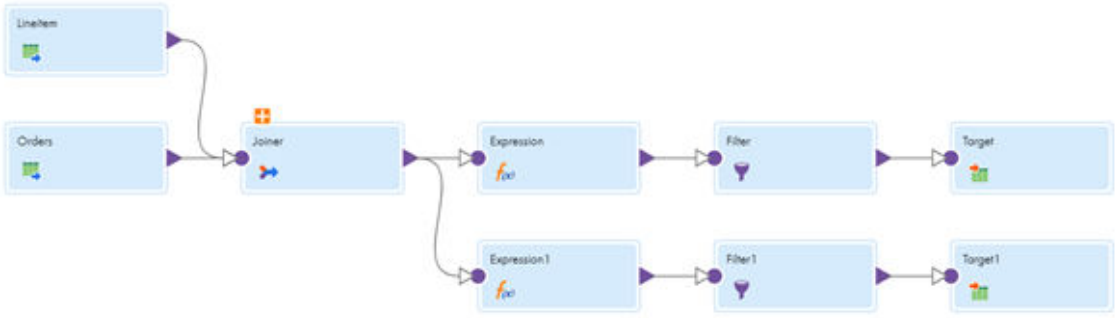


The results show that the mapping performance with pushdown optimization is approximately 46x times faster when compared to the same mapping without pushdown optimization. Full pushdown optimization enabled for mapping that involves reading data from multiple sources that are joined together in Data Integration to write to a single target shows a significant performance gain when compared to the same mapping without pushdown optimization.

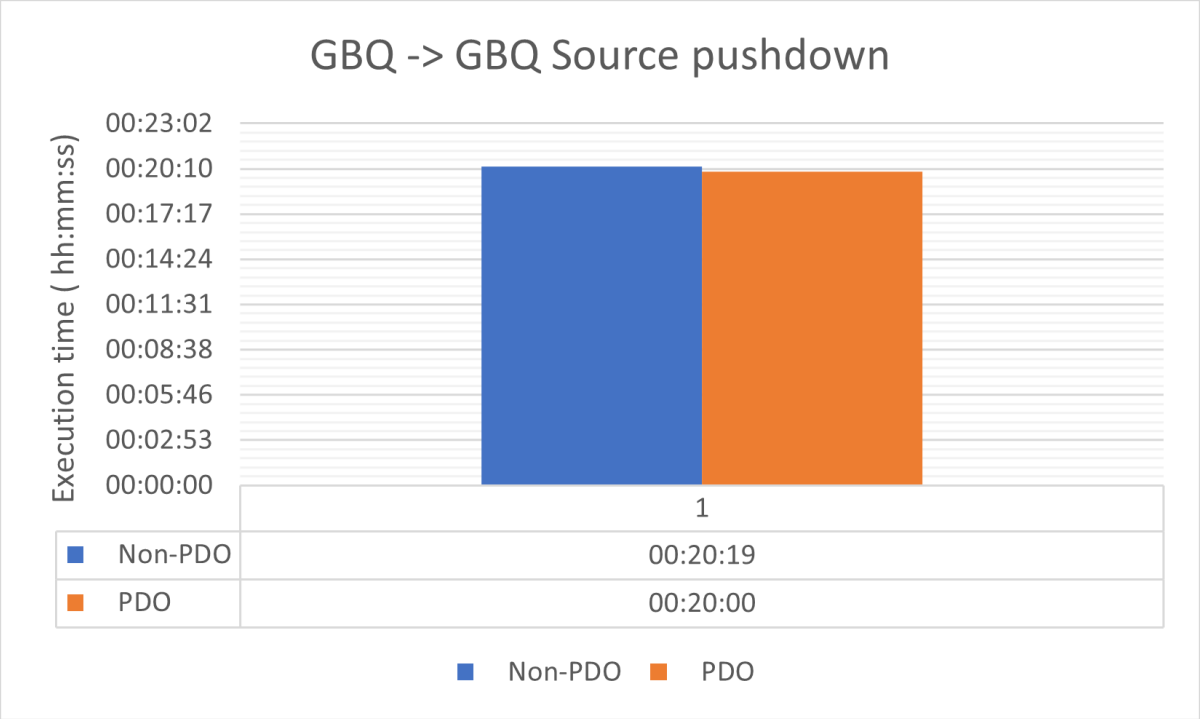
Case 2. Mapping with multiple sources and targets

In this scenario, the mapping includes multiple Google BigQuery sources and multiple Google BigQuery targets. The mapping also includes a Joiner transformation, followed by a series of parallel transformations such as simple expressions and filters. The mapping runs with source pushdown optimization.

The following figure shows the mapping configured with source pushdown optimization:



The following figure shows the results when you run the mapping with and without pushdown optimization:



The results show that the mapping enabled with source pushdown optimization does not provide significant performance gain when compared to the same mapping without pushdown optimization.

Case 3. Mapping with multiple parallel pipelines of transformations followed by a Joiner transformation

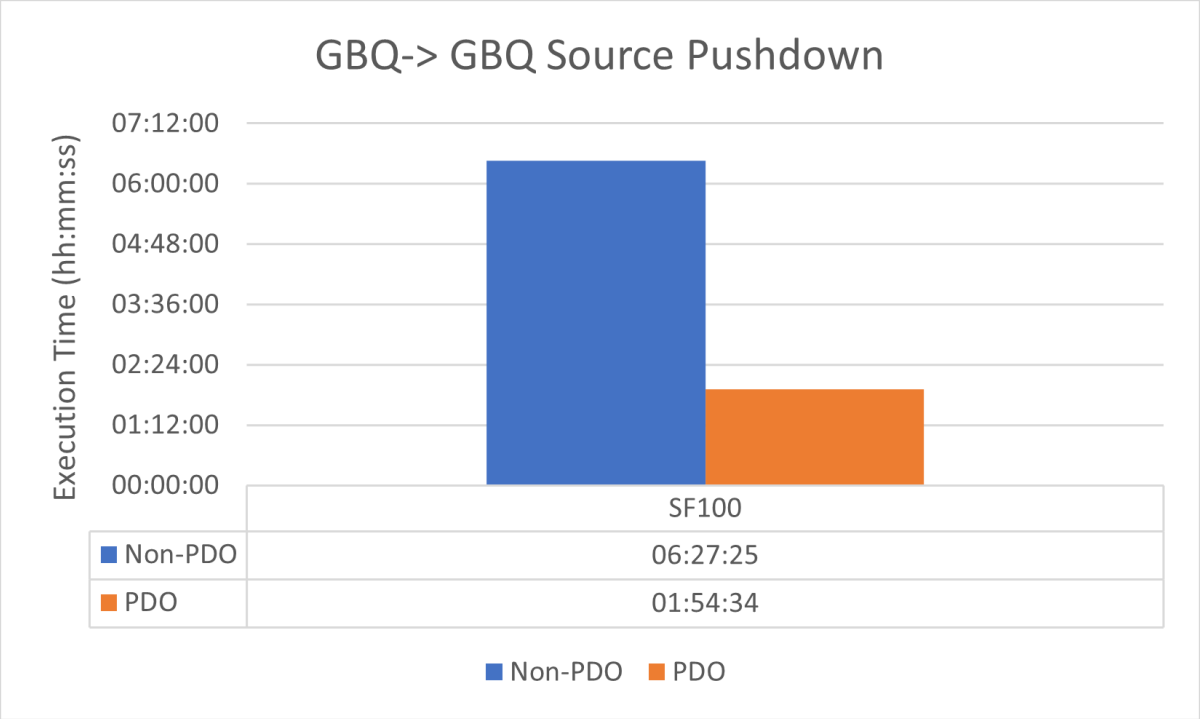
In this scenario, the mapping uses the same transformations as in case 2, except that the Joiner transformation is used further downstream in the mapping.

When the mapping includes multiple parallel pipelines of various transformations followed by a Joiner transformation, optimal performance gain is observed for source pushdown optimization.

The following figure shows the mapping configured with source pushdown optimization:



The following figure shows the results when you run the mapping enabled with and without pushdown optimization:

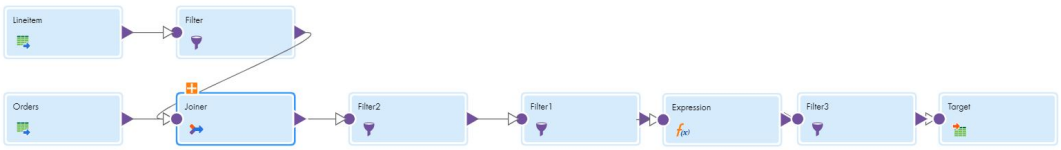


The results show that the mapping enabled with source pushdown optimization recorded approximately 2x times performance gain when compared to the same mapping without pushdown optimization.

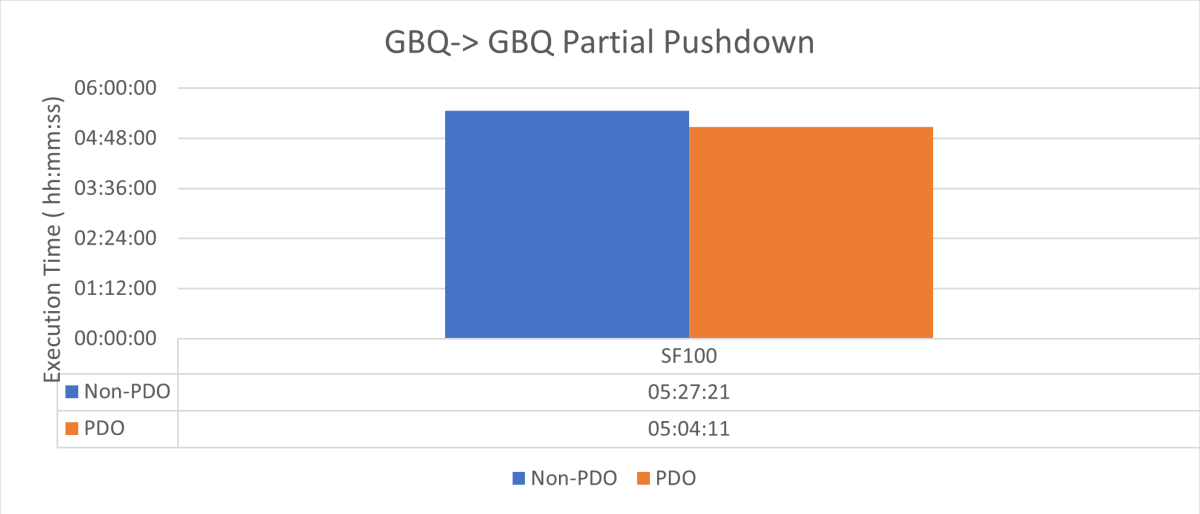
Case 4.Mapping configured with full pushdown optimization runs with partial pushdown optimization

In this scenario, the mapping configured with full pushdown optimization runs with partial pushdown optimization as the Expression transformation contains a function that cannot be pushed to the Google BigQuery database.

The following figure shows the configured mapping with full pushdown optimization:



The following figure shows the results when you run the mapping with and without pushdown optimization:



The results show that there is not much performance gain in this scenario when the mapping runs with full pushdown optimization as compared to a mapping without pushdown optimization.

Performance benefits summary

When you use Google BigQuery V2 Connector, you can observe the following performance benefits while using source or full pushdown optimization in comparison to a mapping without pushdown optimization:

- Full pushdown optimization provides better performance for all supported transformations and operations.
- Source pushdown optimization comparatively does not always provide better performance for all supported transformations and their combinations when compared to a mapping that is not enabled for pushdown optimization.
- Partial pushdown shows negligible performance benefits.

Author

Sonal R.N.